

1 Sodium hydrogencarbonate indicator solution contains sodium hydrogencarbonate.

Some students found out on the internet that this indicator solution changes colour with pH. The students also discovered that this indicator is used to determine the concentration of carbon dioxide in a solution.

The students decided to use this indicator to measure the rate of photosynthesis in algal balls. Algal balls are unicellular algae that are immobilised in alginate. The students assumed that the rate of carbon dioxide uptake was proportional to the rate of photosynthesis.

Fig. 1.1 shows the apparatus that the students used.

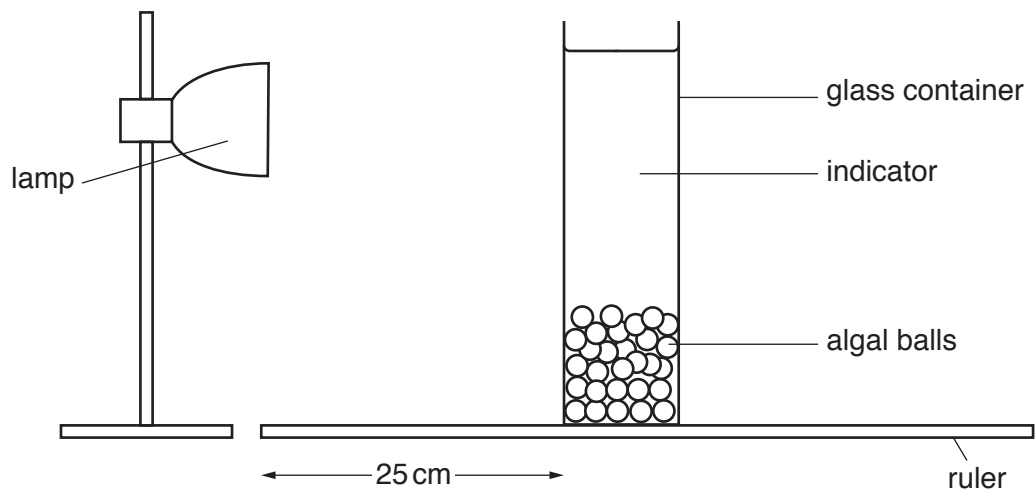


Fig. 1.1

A laboratory technician prepared a series of test-tubes containing hydrogencarbonate indicator solution from pH 7.6 to pH 9.2. The colour of the indicator solution in each test-tube is described in Table 1.1.

Table 1.1

← increasing CO ₂ in indicator solution				decreasing CO ₂ in indicator solution →				
pH 7.6	pH 7.7	pH 8.0	pH 8.2	pH 8.4	pH 8.6	pH 8.8	pH 9.0	pH 9.2
light yellow	dark yellow	light orange	dark orange	red	light magenta	dark magenta	light purple	dark purple

(a) Suggest what was added to each of the test-tubes to maintain the pH.

..... [1]

- (b) Unicellular algae can be immobilised using a technique that is similar to that used to immobilise enzymes.

Outline the method that the students would have used to immobilise the algae.

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..... [3]

- (c) The students used the apparatus shown in Fig. 1.1 and the colours described in Table 1.1 to investigate the effect of light intensity on the rate of photosynthesis.

The students proposed the following hypothesis:

As light intensity increases the rate of photosynthesis increases.

- (i) Identify the independent variable **and** the dependent variable in this investigation.

independent variable

dependent variable

[2]

..... [1]

- (d) The light intensity (I) can be calculated using the formula shown in Fig. 1.2.

$$I = \frac{1}{\text{distance}^2}$$

Fig. 1.2

Use the formula in Fig. 1.2 to calculate the light intensity for the experiment shown in Fig. 1.1.

$$I = \dots\dots\dots [1]$$

The students decided that using a pH probe would give them more accurate data. They decided to measure the pH change for **one** value of light intensity.

Fig. 1.3 shows their results.

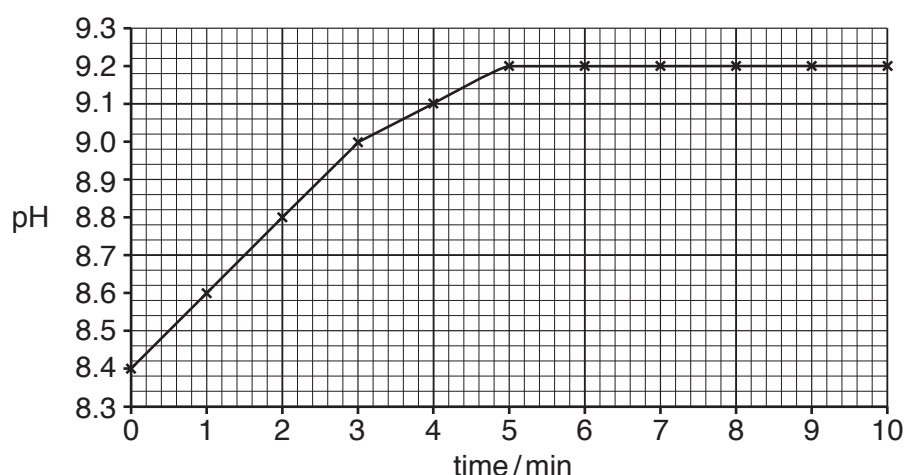


Fig. 1.3

- (e) (i) State why the use of a pH probe provides more accurate data.

..... [1]

- (ii) The data shown in Fig. 1.3 were for a high light intensity.

On Fig. 1.3, sketch the curve that you would expect for a low light intensity. [1]

- (iii) State why carbon dioxide is **not** a limiting factor in this investigation.

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 [1]

- (f) Certain weed killers work by reducing the rate of photosynthesis.

The students modified their method to investigate the effect of three different weed killers on algae at the same light intensity.

The investigation was replicated a number of times and the pH was recorded after 30 minutes. The students calculated the mean pH for each treatment, as shown in Table 1.2.

The students carried out *t*-tests to compare the mean pH for each of the treatments that used weed killers with the treatment that used no weed killer (treatment 1). The students calculated the value of *t* for each of the tests.

Table 1.2 shows the results from their investigation.

Table 1.2

treatment	contents	mean pH after 30 minutes	<i>p</i> value
1	algal balls, indicator and distilled water	9.2	
2	algal balls, indicator and weed killer A	8.5	< 0.001
3	algal balls, indicator and weed killer B	9.0	> 0.05
4	algal balls, indicator and weed killer C	8.7	< 0.05

- (i) Suggest a null hypothesis for comparing the effect of treatment 2 with treatment 1.

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..... [1]

- (ii) The *t*-tests were used to compare the means.

State **one** feature of data that allows use of the *t*-test.

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..... [1]

- (iii) When calculating the effect of weed killer **A** on the rate of photosynthesis, 15 degrees of freedom was chosen. The students carried out 7 replicates of treatment 1.

State how many replicates were carried out for treatment 2.

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..... [1]

- (iv) The students concluded that the weed killers had a significant effect on the rate of photosynthesis.

Explain to what extent the p values in Table 1.2 support their conclusion.

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..... [3]

[Total: 22]